Name: Suyash Singh Tomar CU24250178 roll no:60 labsheet -03

```
import pandas as pd

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-
selectedcolumns.csv")

print(df.head())
print(df.shape)
print(df.columns.tolist())
```

```
   Age Attrition     BusinessTravel  DailyRate              Department
\
0   41       Yes      Travel_Rarely       1102                   Sales

1   49        No  Travel_Frequently        279  Research & Development

2   37       Yes      Travel_Rarely       1373  Research & Development

3   33        No  Travel_Frequently       1392  Research & Development

4   27        No      Travel_Rarely        591  Research & Development


   DistanceFromHome  Education EducationField  EmployeeCount
EmployeeNumber
0                 1          2  Life Sciences              1
1
1                 8          1  Life Sciences              1
2
2                 2          2          Other              1
4
3                 3          4  Life Sciences              1
5
4                 2          1        Medical              1
7
(1470, 10)
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
'EmployeeNumber']
```

```
import numpy as np import
pandas as pd import
matplotlib.pyplot as plt from
scipy import stats import
statsmodels.api as sm
print("Mean:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].mean())
```

```
print("\nMedian:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].median())

print("\nMode:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].mode().iloc[0])

print("\nVariance:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].var())

print("\nStandard Deviation:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].std())
```

```
Mean:
Age                  36.923810
DailyRate           802.485714
DistanceFromHome      9.192517
dtype: float64

Median:
Age                  36.0
DailyRate           802.0
DistanceFromHome      7.0
dtype: float64

Mode:
Age                  35
DailyRate           691
DistanceFromHome      2
Name: 0, dtype: int64

Variance:
Age                     83.455049
DailyRate           162819.593737
DistanceFromHome        65.721251
dtype: float64

Standard Deviation:
Age                  9.135373
DailyRate          403.509100
DistanceFromHome     8.106864
dtype: float64
```

```
print(df[["Age", "DailyRate", "DistanceFromHome"]].describe())
```

```
               Age    DailyRate  DistanceFromHome
count  1470.000000  1470.000000       1470.000000
mean     36.923810   802.485714          9.192517
std       9.135373   403.509100          8.106864
```

```
min      18.000000   102.000000          1.000000
25%      30.000000   465.000000          2.000000
50%      36.000000   802.000000          7.000000
75%      43.000000  1157.000000         14.000000
max      60.000000  1499.000000         29.000000
numeric_df = df[["Age", "DailyRate", "DistanceFromHome"]]
correlation = numeric_df.corr() print(correlation)
                        Age  DailyRate  DistanceFromHome
Age                1.000000   0.010661         -0.001686
DailyRate          0.010661   1.000000         -0.004985
DistanceFromHome -0.001686  -0.004985          1.000000
plt.figure(figsize=(8, 6))

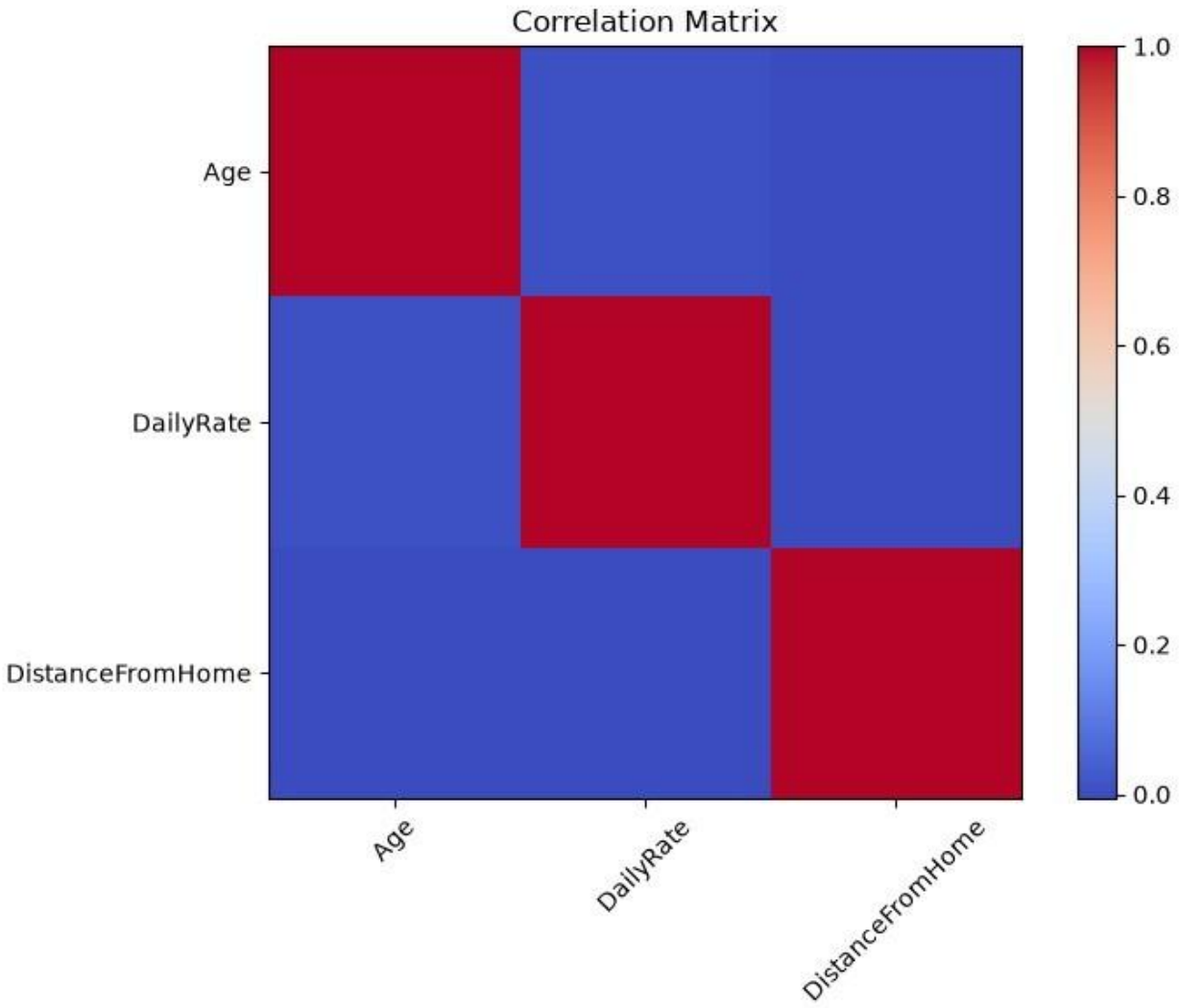
plt.imshow(correlation, cmap="coolwarm")
plt.colorbar()

plt.xticks(
    range(len(correlation.columns)),
correlation.columns,     rotation=45
)

plt.yticks(
    range(len(correlation.columns)),
correlation.columns )

plt.title("Correlation Matrix")

plt.tight_layout()
plt.show()
```




```
r, p = stats.pearsonr(df["Age"], df["DailyRate"])

print("Pearson Correlation:", r) print("P-
value:", p)
Pearson Correlation: 0.01066094264553816
P-value: 0.6829722453880914

left = df[df["Attrition"] == "Yes"]["Age"]

stayed = df[df["Attrition"] == "No"]["Age"]

print("Employees who left:", len(left))
print("Employees who stayed:", len(stayed))
Employees who left: 237

Employees who stayed: 1233
```

```
t_stat, p_value = stats.ttest_ind(
left,     stayed,
    equal_var=False
)

print("T-statistic:", t_stat) print("P-
value:", p_value)
```

```
T-statistic: -5.82801185398895
P-value: 1.3797600649439802e-08
```

```
if p_value < 0.05:
    print("Reject H0")
    print("There is a significant difference in Age between employees
who left and stayed.") else:     print("Fail to reject H0")
    print("There is no significant difference in Age between employees
who left and stayed.")
```

```
Reject H0
There is a significant difference in Age between employees who left
and stayed. print(df["Department"].unique())
```

```
<StringArray>
['Sales', 'Research & Development', 'Human Resources']
Length: 3, dtype: str
```

```
groups = []

for department in df["Department"].unique():
    groups.append(
        df[df["Department"] == department]["Age"]
    )

f_stat, p_value_anova = stats.f_oneway(*groups)

print("F-statistic:", f_stat) print("P-
value:", p_value_anova)
```

```
F-statistic: 0.7655031924976907
P-value: 0.4652855299935297
```

```
if p_value_anova < 0.05:
print("Reject H0")
    print("There is a significant difference in Age among
Departments.") else:     print("Fail to reject H0")
```

```
    print("There is no significant difference in Age among
Departments.")
```

```
Fail to reject H0 There is no significant difference in Age
among Departments.
```

```
X = df["Age"] y =
```

```
df["DailyRate"] X =
```

```
sm.add_constant(X) model =
```

```
sm.OLS(y, X).fit()
```

```
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
========
Dep. Variable:              DailyRate   R-squared:
0.000
Model:                            OLS   Adj. R-squared:
-0.001
Method:                 Least Squares   F-statistic:
0.1669
Date:                Mon, 31 Aug 2026   Prob (F-statistic):
0.683
Time:                        08:31:36   Log-Likelihood:
-10906.
No. Observations:                1470   AIC:
2.182e+04
Df Residuals:                    1468   BIC:
2.183e+04
Df Model:                           1

Covariance Type:            nonrobust
==============================================================================
========
                 coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
--------
const        785.0985     43.847     17.905      0.000     699.089
871.108
Age            0.4709      1.153      0.408      0.683      -1.790
2.732
==============================================================================
========
Omnibus:                     1174.453   Durbin-Watson:
```

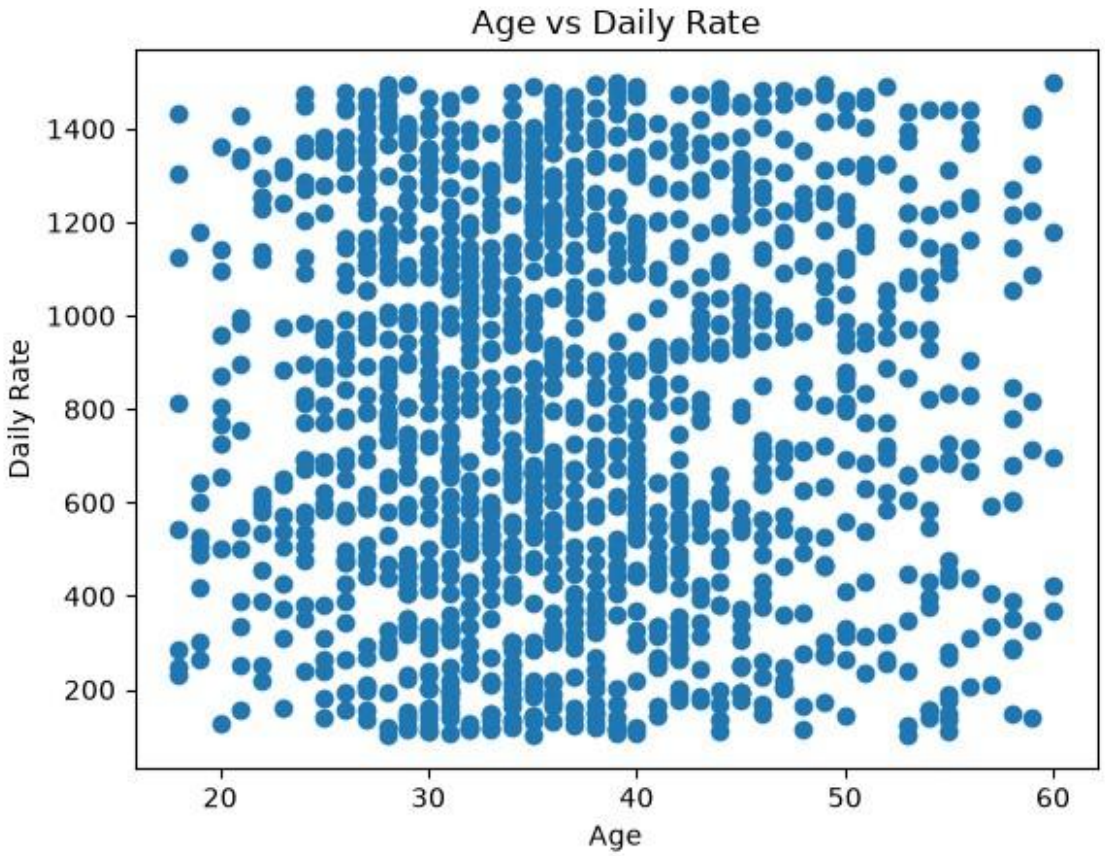
```
2.082
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
88.696
Skew:                          -0.004   Prob(JB):
5.49e-20
Kurtosis:                       1.797   Cond. No.
159.
================================================================================
========

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
```

```
plt.scatter(df["Age"], df["DailyRate"])

plt.xlabel("Age")
plt.ylabel("Daily Rate")
plt.title("Age vs Daily Rate")

plt.show()
```




```
print("R-squared:", model.rsquared)
```

```
R-squared: 0.00011365569809163034
```

```
print("Regression Coefficients:")
print(model.params)
```

```
Regression Coefficients:
const    785.098535
Age        0.470893
dtype: float64
```

```
print("P-values:")
print(model.pvalues)
```

```
P-values: const
5.267571e-65 Age
6.829722e-01 dtype:
float64
```

```
print("Confidence Intervals:")
print(model.conf_int())
```

```
Confidence Intervals:
                0           1
const  699.089362  871.107708
Age     -1.790341    2.732128
print(model.summary())
```

```
                            OLS Regression Results

==============================================================================
========
Dep. Variable:               DailyRate   R-squared:
0.000
Model:                            OLS   Adj. R-squared:
-0.001
Method:                 Least Squares   F-statistic:
0.1669
Date:                Mon, 31 Aug 2026   Prob (F-statistic):
0.683
Time:                        08:31:37   Log-Likelihood:
-10906.
No. Observations:                1470   AIC:
2.182e+04
Df Residuals:                    1468   BIC:
2.183e+04
Df Model:                           1

Covariance Type:            nonrobust
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
const        785.0985     43.847     17.905      0.000     699.089
871.108
Age            0.4709      1.153      0.408      0.683      -1.790
2.732
==============================================================================
Omnibus:                     1174.453   Durbin-Watson:
2.082
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
88.696
Skew:                          -0.004   Prob(JB):
5.49e-20
Kurtosis:                       1.797   Cond. No.
159.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
```

```
print("STATISTICAL ANALYSIS SUMMARY")
print("=" * 40)

print("\n1. T-TEST") print("T-
statistic:", t_stat) print("P-value:",
p_value)

if p_value < 0.05:
    print("Result: Significant difference in Age.")
else:     print("Result: No significant difference in
Age.")

print("\n2. ANOVA") print("F-
statistic:", f_stat) print("P-value:",
p_value_anova)

if p_value_anova < 0.05:     print("Result: Significant difference
among  Departments.")  else:        print("Result:  No  significant
difference among Departments.")

print("\n3. LINEAR REGRESSION") print("R-
squared:", model.rsquared)
```

```
print("Age coefficient:", model.params["Age"])
print("Age p-value:", model.pvalues["Age"])
STATISTICAL ANALYSIS SUMMARY

========================================
1. T-TEST T-statistic: -
5.82801185398895
P-value: 1.3797600649439802e-08
Result: Significant difference in Age.
2. ANOVA

F-statistic: 0.7655031924976907
P-value: 0.4652855299935297

Result: No significant difference among Departments.
3. LINEAR REGRESSION R-squared:
0.00011365569809163034
Age coefficient: 0.47089343162226616
Age p-value: 0.6829722453880929
```

QUESTION AND ANSWER

1. Difference between Descriptive and Inferential Statistics

- Descriptive statistics: Summarizes and describes collected data using mean, median, mode, charts, etc.
- Inferential statistics: Uses sample data to make conclusions or predictions about a larger population.

2. Null Hypothesis and Alternative Hypothesis

- Null Hypothesis (H_0): States that there is no significant difference or relationship.
- Alternative Hypothesis (H_1): States that there is a significant difference or relationship.

Example: H_0: A new medicine has no effect.

H_1: A new medicine has an effect.

3. What is a p-value?

A p-value indicates how likely the observed result is if the null hypothesis is true.

- If p-value < 0.05 → Reject H_0.
- If p-value ≥ 0.05 → Fail to reject H_0.

It helps determine whether a result is statistically significant.

4. Difference between t-test and ANOVA

- t-test: Compares the means of two groups.
- ANOVA: Compares the means of three or more groups.

Example: Comparing marks of two classes → t-test.

Comparing marks of three different classes → ANOVA.

5. Pearson Correlation Coefficient

Pearson correlation measures the strength and direction of a linear relationship between two variables.
Its values range from -1 to +1:

- +1: Perfect positive correlation
- 0: No linear correlation
- -1: Perfect negative correlation

6. Significance of R^2 in Linear Regression

R^2 (Coefficient of Determination) shows how much variation in the dependent variable is explained by the independent variable(s).
For example, $R^2 = 0.80$ means 80% of the variation is explained by the regression model.

7. Importance of Statistical Analysis before Machine Learning

Statistical analysis helps to:

- Understand the dataset.
- Detect outliers and missing values.
- Identify relationships between variables.
- Select useful features.
- Check assumptions and data quality.

This can improve the performance and reliability of ML models.

8. Assumptions before t-test or ANOVA

Important assumptions include:

- Data should be approximately normally distributed.
- Observations should be independent.
- Groups should have approximately equal variances (for standard versions of these tests).
- The dependent variable should generally be numerical.

9. Regression Analysis in Forecasting and Decision-Making

Regression identifies relationships between variables and can be used to predict future outcomes.
For example, a company can use past sales and advertising expenses to predict future sales and decide how much to spend on marketing.

10. Two Real-World Applications of Hypothesis Testing

1. **Medicine:** Testing whether a new drug is more effective than an existing treatment.
2. **Business:** Testing whether a new advertising campaign significantly increases sales.